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**Course:** Calculus Beta

**Assignment:** MyLab Math - Uge 6

Find the Taylor polynomials of orders 0, 1, 2, and 3 generated by  $f$  at  $x = a$ .

$$f(x) = \frac{6}{x}, a = 2$$

The partial sum  $P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$  is the Taylor polynomial of order  $n$  for  $f$  at  $x = a$ .  $f^{(k)}$  represents the  $k$ th derivative of the function  $f(x)$  and  $f^{(0)}(x)$  represents  $f(x)$ .

To find the 0 order Taylor polynomial for  $f(x) = \frac{6}{x}$ , substitute 0 for  $k$  in  $\frac{f^{(k)}(a)}{k!} (x-a)^k$ . This simplifies to  $f(a)$ . Recall that  $0! = 1$ .

$$\begin{aligned} P_0(x) &= f(a) \\ &= \frac{6}{(2)} \\ &= 3 \end{aligned}$$

The Taylor polynomial of order 1 for  $f(x) = \frac{6}{x}$  can be found by evaluating  $\frac{f^{(k)}(a)}{k!} (x-a)^k$  at  $k = 1$  and adding the Taylor polynomial found for order 0.

$$3 + f'(a)(x-a)$$

Find the derivative of  $f(x) = \frac{6}{x}$  in order to find  $f'(a)$ .

$$\frac{d}{dx} \left( \frac{6}{x} \right) = -\frac{6}{x^2}$$

With  $f'(x) = -\frac{6}{x^2}$ , find  $f'(a)(x-a)$ .

$$f'(a)(x-a) = -\frac{6}{(2)^2} (x-(2)) = -\frac{3}{2} (x-2)$$

Thus, the Taylor polynomial of order 1 for  $f(x) = \frac{6}{x}$  is  $P_1(x) = 3 - \frac{3}{2}(x-2)$ .

The Taylor polynomial of order 2 for  $f(x) = \frac{6}{x}$  can be found by evaluating  $\frac{f^{(k)}(a)}{k!} (x-a)^k$  at  $k = 2$  and adding the Taylor polynomial found for order 1.

$$3 - \frac{3}{2}(x-2) + \frac{f''(a)}{2!} (x-a)^2$$

Find the derivative of  $f'(x)$  in order to find  $f''(a)$ .

$$\frac{d}{dx} \left( -\frac{6}{x^2} \right) = \frac{12}{x^3}$$

With  $f''(x) = \frac{12}{x^3}$ , find  $\frac{f''(a)}{2!} (x-a)^2$ .

$$\frac{f''(a)}{2!} (x-a)^2 = \frac{12}{2!(2)^3} (x-(2))^2 = \frac{3}{4} (x-2)^2$$

Thus, the Taylor polynomial of order 2 for  $f(x) = \frac{6}{x}$  is  $P_2(x) = 3 - \frac{3}{2}(x-2) + \frac{3}{4}(x-2)^2$ .

The Taylor polynomial of order 3 for  $f(x) = \frac{6}{x}$  can be found by evaluating  $\frac{f^{(k)}(a)}{k!} (x-a)^k$  at  $k = 3$  and adding the Taylor polynomial found for order 2.

$$3 - \frac{3}{2}(x-2) + \frac{3}{4}(x-2)^2 + \frac{f'''(a)}{3!}(x-a)^3$$

Find the derivative of  $f''(x)$  in order to find  $f'''(a)$ .

$$\frac{d}{dx} \left( \frac{12}{x^3} \right) = -\frac{36}{x^4}$$

With  $f'''(x) = -\frac{36}{x^4}$ , find  $\frac{f'''(a)}{3!}(x-a)^3$ .

$$\frac{f'''(a)}{3!}(x-a)^3 = -\frac{36}{3!(2)^4}(x-(2))^3 = -\frac{3}{8}(x-2)^3$$

Thus, the Taylor polynomial of order 3 for  $f(x) = \frac{6}{x}$  is  $P_3(x) = 3 - \frac{3}{2}(x-2) + \frac{3}{4}(x-2)^2 - \frac{3}{8}(x-2)^3$ .